Machine Learning A (2026–2027)
Home Assignment 3

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1 Optimal Classification

Question 1

Define the set of possible y-classes $\mathcal{C}_k \in \{0, 1\}$. The optimal deterministic classification strategy is the one that maximizes the probability of succes given x:

$$h(x) = \arg\max_{\mathcal{C}_k} \mathbb{P}\left(Y = \mathcal{C}_k \mid X = x\right) = \begin{cases} 0 & \text{, if } x = 0 \\ 0 & \text{, if } x = 1 \end{cases}.$$

The risk is calculated where we use the law of total probability and zero-one loss

$$\begin{aligned}
\mathbb{E}\left[\ell\left(y, h(x)\right)\right] &= \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}\left(Y = y, X = x\right) \ell\left(y, h(x)\right) \\
&= \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(X = x) \mathbb{P}\left(Y = y \mid X = x\right) \ell\left(y, h(x)\right)
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}(X = 0) \mathbb{P}\left(Y = 0 \mid X = 0\right) \ell\left(0, h(0)\right) \\
&+ \mathbb{P}(X = 0) \mathbb{P}\left(Y = 1 \mid X = 0\right) \ell\left(1, h(0)\right) \\
&+ \mathbb{P}(X = 1) \mathbb{P}\left(Y = 0 \mid X = 1\right) \ell\left(0, h(1)\right) \\
&+ \mathbb{P}(X = 1) \mathbb{P}\left(Y = 1 \mid X = 1\right) \ell\left(1, h(1)\right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \cdot 1 \cdot 0 + \frac{1}{2} \cdot 0 \cdot 1 + \frac{1}{2} \cdot 0.8 \cdot 0 + \frac{1}{2} \cdot 0.2 \cdot 1 \\
&= \frac{1}{10}.
\end{aligned}$$

Question 2

Using the fact that Y is conditionally independent of our hypothesis h' and that $h'(x) \equiv \mathbb{P}\left(h'(x) = \hat{y} \mid X = x\right) = \mathbb{P}\left(Y = \hat{y} \mid X = x\right)$ we calculate the risk of the hypothesis

$$\begin{aligned}
\mathbb{E}\left[\ell\left(y, h'(x)\right)\right] &= \sum_{\hat{y}\in\{0,1\}} \sum_{x\in\{0,1\}} \sum_{y\in\{0,1\}} \mathbb{P}\left(Y = y, X = x, h'(x) = \hat{y}\right) \ell\left(y, \hat{y}\right) \\
&= \sum_{\hat{y}\in\{0,1\}} \sum_{x\in\{0,1\}} \sum_{y\in\{0,1\}} \mathbb{P}\left(Y = y, h'(x) = \hat{y} \mid X = x\right) \mathbb{P}\left(X = x\right) \ell\left(y, \hat{y}\right) \\
&= \sum_{\hat{y}\in\{0,1\}} \sum_{x\in\{0,1\}} \sum_{y\in\{0,1\}} \mathbb{P}\left(Y = y \mid X = x\right) \mathbb{P}\left(h'(x) = \hat{y} \mid X = x\right) \mathbb{P}\left(X = x\right) \ell\left(y, \hat{y}\right) \\
&= \sum_{\hat{y}\in\{0,1\}} \sum_{x\in\{0,1\}} \sum_{y\in\{0,1\}} \mathbb{P}\left(Y = y \mid X = x\right) \mathbb{P}\left(Y = \hat{y} \mid X = x\right) \mathbb{P}\left(X = x\right) \ell\left(y, \hat{y}\right)
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1)\ell(1,1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(X = 1)\ell(1,0) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0)\ell(1,1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(X = 0)\ell(1,0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1)\ell(0,1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(X = 1)\ell(0,0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0)\ell(0,1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(X = 0)\ell(0,0) \\
&= \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(X = 1) \\
&\quad + \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(X = 0) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 1)\mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1) \\
&\quad + \mathbb{P}(Y = 0 \mid X = 0)\mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0) \\
&= \frac{1}{5}\frac{4}{5}\frac{1}{2} + 0 \cdot 1 \cdot \frac{1}{2} + \frac{4}{5}\frac{1}{5}\frac{1}{2} + 1 \cdot 0 \cdot \frac{1}{2} \\
&= \frac{4}{25}.
\end{aligned}$$

So in this case the first hypothesis

$$h(x) = \arg\max_{\mathcal{C}_k} \mathbb{P}\left(Y = \mathcal{C}_k \mid X = x\right) = \begin{cases} 0 & \text{, if } x = 0 \\ 0 & \text{, if } x = 1 \end{cases},$$

is the optimal.

Question 3

Let the observed input be $X = x$. Then say $\mathbb{P}(h(x) = 1 \mid X = x) = \gamma_x$ and $\mathbb{P}(h(x) = 0 \mid X = x) = 1 - \gamma_x$ then the expected loss is

$$\begin{aligned}
\mathbb{E}[\ell(y, h(x)) \mid X = x] &= \sum_{\hat{y} \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, h(x) = \hat{y} \mid X = x)\, \ell(y, \hat{y}) \\
&= \sum_{\hat{y} \in \{0,1\}} \sum_{y \in \{0,1\}} \mathbb{P}(Y = y \mid X = x)\, \mathbb{P}(h(x) = \hat{y} \mid X = x)\, \ell(y, \hat{y})
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}(Y = 1 \mid X = x)\, \mathbb{P}(h(0) = 1 \mid X = x)\, \ell(1,1) \\
&+ \mathbb{P}(Y = 1 \mid X = x)\, \mathbb{P}(h(0) = 0 \mid X = x)\, \ell(1,0) \\
&+ \mathbb{P}(Y = 0 \mid X = x)\, \mathbb{P}(h(0) = 1 \mid X = x)\, \ell(0,1) \\
&+ \mathbb{P}(Y = 0 \mid X = x)\, \mathbb{P}(h(0) = 0 \mid X = x)\, \ell(0,0)
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}(Y = 1 \mid X = x)\, \mathbb{P}(h(0) = 0 \mid X = x) \\
&+ \mathbb{P}(Y = 0 \mid X = x)\, \mathbb{P}(h(0) = 1 \mid X = x)
\end{aligned}$$

$$= \gamma_x \mathbb{P}(Y = 1 \mid X = x) + (1 - \gamma_x) \mathbb{P}(Y = 0 \mid X = x),$$

where the first equal sign uses the fact that Y is independent of our hypothesis h. Using the above we calculate the risk

$$\begin{aligned}
\mathbb{E}[\ell(y, h(x))] &= \mathbb{E}[\mathbb{E}[\ell(y, h(x)) \mid X = x]] \\
&= \sum_{x \in \{0,1\}} \mathbb{P}(X = x)\, \mathbb{E}[\ell(y, h(x)) \mid X = x] \\
&= \frac{1}{2}(\mathbb{E}[\ell(y, h(x)) \mid X = 0]) + \frac{1}{2}(\mathbb{E}[\ell(y, h(x)) \mid X = 1])
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2}(\gamma_0 \mathbb{P}(Y = 1 \mid X = 0) + (1 - \gamma_0) \mathbb{P}(Y = 0 \mid X = 0)) \\
&+ \frac{1}{2}(\gamma_1 \mathbb{P}(Y = 1 \mid X = 1) + (1 - \gamma_1) \mathbb{P}(Y = 0 \mid X = 1))
\end{aligned}$$

Because this is linear in the probabilities the answer is immediately apparent. When minimizing w.r.t. γ_0 and γ_1 the optimal strategy will deterministically choose the output that is most likely given $X = x$, in most cases. If the probabilities of $Y = 0$ and $Y = 1$ are equal then the weights γ_x does not matter. Formally:

$$\gamma_x^* = \mathbb{P}\left(h(x) = 1 \mid X = x\right) = \begin{cases} 1 & \text{, if } \mathbb{P}\left(Y = 1 \mid X = x\right) > \mathbb{P}\left(Y = 0 \mid X = x\right) \\ 0 & \text{, if } \mathbb{P}\left(Y = 1 \mid X = x\right) < \mathbb{P}\left(Y = 0 \mid X = x\right) \\ [0,1] & \text{, if } \mathbb{P}\left(Y = 1 \mid X = x\right) = \mathbb{P}\left(Y = 0 \mid X = x\right) \end{cases}$$

2 Logistic Regression

2.1 Cross-entropy error measure

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

Firstly I rewrite $-\ln\left[\mathbb{P}\left(Y = y_n \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right]$ where I use the fact that the sample space of the outcome variable is $Y \in \{-1, 1\}$.

$$\begin{aligned} -\ln\left[\mathbb{P}\left(Y = y_n \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right] &= \mathbb{I}_{(y_n=1)}\left\{-\ln\left[\mathbb{P}\left(Y = 1 \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right]\right\} \\ &\quad + \mathbb{I}_{(y_n=-1)}\left\{-\ln\left[\mathbb{P}\left(Y = -1 \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right]\right\} \\ &= \mathbb{I}_{(y_n=1)}\left\{-\ln\left[\mathbb{P}\left(Y = 1 \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right]\right\} \\ &\quad + \mathbb{I}_{(y_n=-1)}\left\{-\ln\left[1 - \mathbb{P}\left(Y = 1 \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right]\right\} \\ &= \mathbb{I}_{(y_n=1)}\left\{-\ln\left[h(\boldsymbol{x}_n)\right]\right\} + \mathbb{I}_{(y_n=-1)}\left\{-\ln\left[1 - h(\boldsymbol{x}_n)\right]\right\} \\ &= \mathbb{I}_{(y_n=1)} \ln\left[\frac{1}{h(\boldsymbol{x}_n)}\right] + \mathbb{I}_{(y_n=-1)} \ln\left[\frac{1}{1 - h(\boldsymbol{x}_n)}\right] \end{aligned}$$

where $\mathbb{I}_{(.)}$ is the indicator function. Inserting this into the general negative logarithmic likelihood function yields the wanted result:

$$\begin{aligned} \mathcal{L}(\boldsymbol{w}) &= -\frac{1}{N}\sum_{n=1}^{N} \ln\left[\mathbb{P}\left(Y = y_n \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right] \\ &= \frac{1}{N}\sum_{n=1}^{N} -\ln\left[\mathbb{P}\left(Y = y_n \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right] \\ &= \frac{1}{N}\sum_{n=1}^{N}\left[\mathbb{I}_{(y_n=1)} \ln\left[\frac{1}{h(\boldsymbol{x}_n)}\right] + \mathbb{I}_{(y_n=-1)} \ln\left[\frac{1}{1 - h(\boldsymbol{x}_n)}\right]\right]. \end{aligned}$$

2.1.2 *Cross-entropy and logistic regression*

First we note the following property of the sigmoid function:

$$\begin{aligned}\theta(s) &= \frac{1}{1+\exp\{-s\}} \implies \\ 1-\theta(s) &= \frac{\exp\{-s\}}{1+\exp\{-s\}} = \frac{1}{1+\exp\{s\}} = \theta(-s).\end{aligned}$$

Using this and $y_n \in \{-1, 1\}$ we simply rewrite eq. (1) from the assignment:

$$\begin{aligned}\mathcal{L}(\boldsymbol{w}) &= \frac{1}{N}\sum_{n=1}^{N} \ln\left[1+\exp\left\{-y_n \boldsymbol{w}^\top \boldsymbol{x}\right\}\right] \\ &= -\frac{1}{N}\sum_{n=1}^{N}\left[\mathbb{I}_{(y_n=1)}\left(1+\exp\left\{-\boldsymbol{w}^\top \boldsymbol{x}\right\}\right) + \mathbb{I}_{(y_n=-1)}\left(1+\exp\left\{\boldsymbol{w}^\top \boldsymbol{x}\right\}\right)\right] \\ &= \frac{1}{N}\sum_{n=1}^{N}\left[\mathbb{I}_{(y_n=1)}\ln\left[\frac{1}{h(\boldsymbol{x}_n)}\right] + \mathbb{I}_{(y_n=-1)}\ln\left[\frac{1}{1-h(\boldsymbol{x}_n)}\right]\right].\end{aligned}$$

Yielding the wanted result.

2.2 Logistic regression loss gradient

2.2.1 *Gradient for* -1 *and* 1

Firstly we define a useful function $l_n(\cdot) \equiv \ln\left(1+\exp\left\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\right\}\right)$ as the likelihood contribution of observation n such that

$$\begin{aligned}\mathcal{L}(\boldsymbol{w}) &= \frac{1}{N}\sum_{n=1}^{N}\ln\left(1+\exp\left\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\right\}\right) \\ &= \frac{1}{N}\sum_{n=1}^{N} l_n\left(\boldsymbol{w}\right),\end{aligned}$$

By the summation rule of differentiation we have

$$\nabla\mathcal{L}(\boldsymbol{w}) = \frac{1}{N}\sum_{n=1}^{N}\nabla l_n(\boldsymbol{w}), \tag{2.1}$$

and by calculating an arbitrary of these N terms yields

$$\begin{aligned}
\nabla l_n(\boldsymbol{w}) &= \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial w_1} & \frac{\partial \mathcal{L}}{\partial w_2} & \cdots & \frac{\partial \mathcal{L}}{\partial w_d} \end{bmatrix}^\top \\
&= \begin{bmatrix} \frac{1}{1+\exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}} \exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\} (-y_n x_{n1}) \\ \frac{1}{1+\exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}} \exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\} (-y_n x_{n2}) \\ \vdots \\ \frac{1}{1+\exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}} \exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\} (-y_n x_{nd}) \end{bmatrix} \\
&= -y_n \frac{\exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}}{1+\exp\{-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}} \begin{bmatrix} x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}^\top \\
&= -\frac{y_n \boldsymbol{x}_n}{1+\exp\{y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}},
\end{aligned}$$

which we insert into (2.1) to obtain the wanted result

$$\begin{aligned}
\mathcal{L}(\boldsymbol{w}) &= -\frac{1}{N} \sum_{n=1}^{N} \frac{y_n \boldsymbol{x}_n}{1+\exp\{y_n \boldsymbol{w}^\top \boldsymbol{x}_n\}} \\
&= \frac{1}{N} \sum_{n=1}^{N} -y_n \boldsymbol{x}_n \frac{1}{1+\exp\{-(-y_n \boldsymbol{w}^\top \boldsymbol{x}_n)\}} \\
&= \frac{1}{N} \sum_{n=1}^{N} -y_n \boldsymbol{x}_n \theta\left(-y_n \boldsymbol{w}^\top \boldsymbol{x}_n\right).
\end{aligned}$$

2.2.2 *Gradient for* 0 *and* 1

The exact same procedure is completed for the new encoding of the probability function:

$$\begin{aligned}
\mathbb{P}\left(Y = y_n \mid X = \boldsymbol{x}_n\right) &= \begin{cases} \theta(\boldsymbol{w}^\top \boldsymbol{x}) & \text{, if } y = 1 \\ \theta(-\boldsymbol{w}^\top \boldsymbol{x}) & \text{, if } y = 0 \end{cases} \\
&= \left[\theta(\boldsymbol{w}^\top \boldsymbol{x})\right]^{y_n} \left[\theta(-\boldsymbol{w}^\top \boldsymbol{x})\right]^{1-y_n}.
\end{aligned}$$

That is, we again setup the corresponding negative logarithmic likelihood function

$$\begin{aligned}\mathcal{L}(\boldsymbol{w}) &= -\frac{1}{N}\sum_{n=1}^{N}\ln\left[\mathbb{P}\left(Y = y_n \mid X = \boldsymbol{x}_n; \boldsymbol{w}\right)\right] \\ &= -\frac{1}{N}\sum_{n=1}^{N}\left\{\ln\left[\theta(\boldsymbol{w}^\top\boldsymbol{x})^{y_n}\theta(-\boldsymbol{w}^\top\boldsymbol{x})^{1-y_n}\right]\right\} \\ &= -\frac{1}{N}\sum_{n=1}^{N}\left\{y_n\ln\left[\theta(\boldsymbol{w}^\top\boldsymbol{x})\right] + (1-y_n)\ln\left[\theta(-\boldsymbol{w}^\top\boldsymbol{x})\right]\right\},\end{aligned}$$

and calculate the gradient

$$\begin{aligned}\nabla\mathcal{L}(\boldsymbol{w}) &= -\frac{1}{N}\sum_{n=1}^{N}\begin{bmatrix} y_n\frac{1}{1+\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}}\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}(-x_{n1}) + (1-y_n)\frac{1}{1+\exp\{\boldsymbol{w}^\top\boldsymbol{x}\}}\exp\{\boldsymbol{w}^\top\boldsymbol{x}\}x_{n1} \\ y_n\frac{1}{1+\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}}\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}(-x_{n2}) + (1-y_n)\frac{1}{1+\exp\{\boldsymbol{w}^\top\boldsymbol{x}\}}\exp\{\boldsymbol{w}^\top\boldsymbol{x}\}x_{n2} \\ \vdots \\ y_n\frac{1}{1+\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}}\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}(-x_{nd}) + (1-y_n)\frac{1}{1+\exp\{\boldsymbol{w}^\top\boldsymbol{x}\}}\exp\{\boldsymbol{w}^\top\boldsymbol{x}\}x_{nd}\end{bmatrix} \\ &= -\frac{1}{N}\sum_{n=1}^{N}\left[\frac{1}{1+\exp\left\{-\boldsymbol{w}^\top\boldsymbol{x}\right\}}\right]\begin{bmatrix}\left(-\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}y_n + (1-y_n)\right)x_{n1} \\ \left(-\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}y_n + (1-y_n)\right)x_{n2} \\ \vdots \\ \left(-\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\}y_n + (1-y_n)\right)x_{nd}\end{bmatrix}\end{aligned}$$

$$\begin{aligned}&= -\frac{1}{N}\sum_{n=1}^{N}\left[\frac{1}{1+\exp\left\{-\boldsymbol{w}^\top\boldsymbol{x}\right\}}\right]\Big(y_n(1-\exp\{-\boldsymbol{w}^\top\boldsymbol{x}\})+1\Big)\,\boldsymbol{x}_n \\ &= -\frac{1}{N}\sum_{n=1}^{N}\left[y_n + \frac{1}{1+\exp\left\{-\boldsymbol{w}^\top\boldsymbol{x}\right\}}\right]\boldsymbol{x}_n \\ &= -\frac{1}{N}\sum_{n=1}^{N}\Big[y_n + \theta(\boldsymbol{w}^\top\boldsymbol{x})\Big]\,\boldsymbol{x}_n.\end{aligned}$$

Done!

2.2.3 Influence of misclassified examples

For the encoding $y_n \in \{-1, 1\}$, the contribution of observation n to the gradient is

$$-y_n x_n \theta(-y_n w^\top x_n).$$

An observation is correctly classified if $y_n w^\top x_n > 0$ and misclassified if $y_n w^\top x_n < 0$. Therefore,

$$\begin{cases} y_n w^\top x_n > 0 & \Rightarrow \theta(-y_n w^\top x_n) < \frac{1}{2}, \\ y_n w^\top x_n < 0 & \Rightarrow \theta(-y_n w^\top x_n) > \frac{1}{2}. \end{cases}$$

Hence, relative to the size of x_n, a misclassified observation contributes more to the gradient.

2.3 Log-odds

By direct calculation we get

$$\begin{aligned} \boldsymbol{w}^\top \boldsymbol{x} + b = \ln\left[\frac{\mathbb{P}\left(Y=1 \mid X=x\right)}{\mathbb{P}\left(Y=0 \mid X=x\right)}\right] &\iff \\ -(\boldsymbol{w}^\top \boldsymbol{x} + b) = \ln\left[\frac{\mathbb{P}\left(Y=0 \mid X=x\right)}{\mathbb{P}\left(Y=1 \mid X=x\right)}\right] &\iff \\ \mathbb{P}\left(Y=1 \mid X=x\right)\exp\left\{-(\boldsymbol{w}^\top \boldsymbol{x} + b)\right\} = \mathbb{P}\left(Y=0 \mid X=x\right) = \left(1 - \mathbb{P}\left(Y=1 \mid X=x\right)\right) &\iff \\ 1 = \mathbb{P}\left(Y=1 \mid X=x\right)\left(1 + \exp\left\{-(\boldsymbol{w}^\top \boldsymbol{x} + b)\right\}\right) &\iff \\ \mathbb{P}\left(Y=1 \mid X=x\right) = \frac{1}{1+\exp\left\{-(\boldsymbol{w}^\top \boldsymbol{x} + b)\right\}} = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b). \end{aligned}$$

That is if $\boldsymbol{w}^\top \boldsymbol{x} + b$ encodes the relative probability on log scale of observing $Y = 1$ relative to $Y = 0$ given $X = x$, then the sigmoid function describes the probability of observing $Y = 1$.

3 Sleep Well

3.1 Data understanding and preprocessing

I am using python 3 and the numpy package to calculate the shares of observations in each of the five labeling categories.[1] The result is displayed in tabel 1. Notably the labels 0 and 2 are much more prevalent in the data and we therefore need to admit for sufficiently complex classification models to partition the data appropriately.

[1] For the rest of part 3 i will use sci-kit learn package for the implementation of the machine learning algorithms.

Category	Share
0	0.521
1	0.096
2	0.253
3	0.047
4	0.084

Table 1: Category shares

Note: The table displays the shares of observations labeled as each category in the training data.

3.2 Classification

3.2.1 Logistic regression

To fit the multimodal logistic regression I use the ski-cit learn (sklearn) implementation. Table 2 shows the resulting in sample and out of sample empirical risk calculations. Curiously we note that the out of sample empirical risk is smaller, which will be the case for all the fittet models in this section.

Sample type	Risk
In Sample	0.149
Out of Sample	0.099

Table 2: Logistic regression empirical risk

Note: Risk from fitted multimodal logistic regression.
Datasource: https://github.com/christian-igel/ML/tree/main/data/Sleep/

3.2.2 Random forest

Training the random forest is done using the sklearn implementation called RandomForestClassifier. For all the models i run a 5 fold cross validation using sklearns GridSearchCV module to choose the max depth and minimum number of samples in the trees leafs. [2] For all forest sizes the maxmimum depth and min samples in leaf that minimizes the CV risk is 10 and 1 respectively. The results are shown in table 3, where the out of bag error (OOB error) is calculated by utilizing that any given training data-point is not used in about one third of the trees.

[2] I first shuffle the data (before applying CV) using the integrated StratifiedKFold also from sklearn.

Sample type	Risk (50 trees)	Risk (100 trees)	Risk (200 trees)
In Sample	0.107	0.107	0.106
OOB error	0.147	0.146	0.146
Out of Sample	0.106	0.106	0.105

Table 3: Random forest empirical risk

Note: Risk and OOB error from trained random forest with 50, 100 and 200 trees respectively. The max depth and minimum of samples in the forests leafs are 10 and 1 for all the models. Datasource: https://github.com/christian-igel/ML/tree/

3.2.3 Nearest neighbor

The model selection (choosing K) is again done using shuffled 5 fold CV as in 3.2.2 and using sklearns KNeighborsClassifier. The K minimizing the CV risk in the data is $K = 24$ and the risk estimates for this model is presented in table 4.

Sample type	Risk
In Sample	0.142
Out of Sample	0.102

Table 4: K-NN empirical risk

Note: Risk from K-NN (K=24) model fitted using shuffled 5 fold CV.
Datasource: https://github.com/christian-igel/ML/tree/